SnapLex Technical Architecture & Engineering Spec

1. System Architecture

UI Layer (Floating Window)

Capture Layer (Screenshot + Overlay)

OCR Layer (PaddleOCR/Tesseract)

Clipboard Layer

Translation Layer (Providers)

Storage Layer (Config + History)

2. Tech Stack

Python + PySide6 (MVP)

PaddleOCR

mss/pyautogui

LibreTranslate/OpenAI/DeepL APIs

PyInstaller packaging

3. Module Design

capture_service: screen grab + region select

ocr_service: image to text

clipboard_service: clipboard access

translation_service: provider abstraction

4. Translation Provider Interface

translate(text, source_lang, target_lang) -> translated_text

5. Runtime Flow

User Action -> Capture -> OCR/Clipboard -> Normalize -> Translate -> Render Popup

6. Error Handling

Clipboard empty fallback

OCR failure retry

Network timeout fallback to local provider

7. Performance Considerations

Asynchronous OCR

Caching recent translations

Lazy loading OCR models